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CSC-580 Fall 2025
Midterm Exam Submission

1. Zachman:

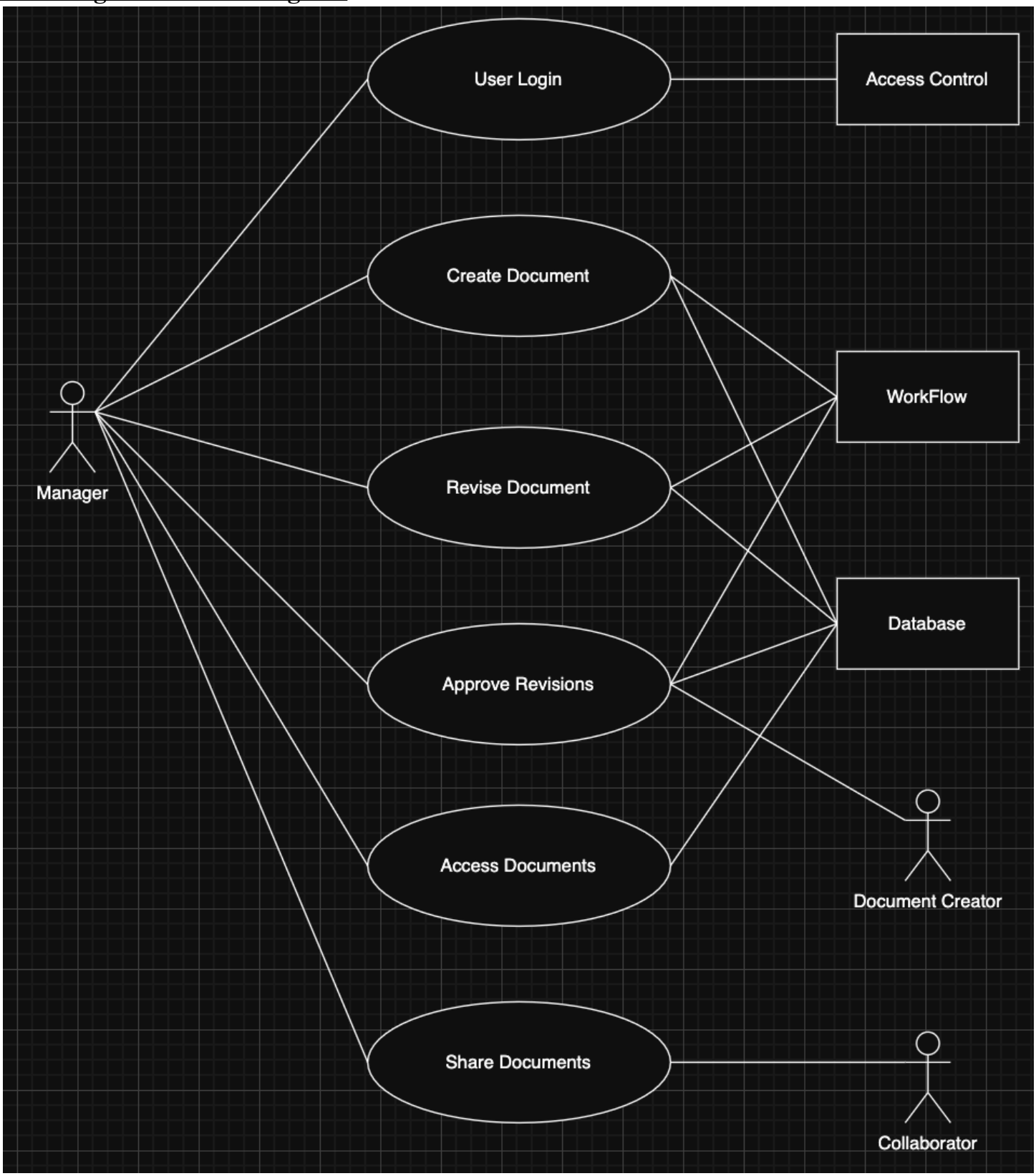
Scope Perspective	What Context (Data)
(Executive's View)	1. Documents
	2. Revisions
	3. Authorizations / Permissions
	4. Metadata
	5. workflows
	How Context (Function)
	1. Creating and storing documents
	2. Document Retrieval
	3. Document Revisioning
	4. Document Access Limitiation
	5. Communication and Collaboration
	Where Context (Location)
	1. Headquarters
	2. Satellite locations (buildings other than headquarters)
	3. Supplier Locations
	4. Remote Work areas
	5. International Locations
	Who Context (Users/Stakesholders)
	1. Document Owner (Use of System)
	2. Managers (Broad overview of System)
	3. Quality Assurance Personnel (Revisioning)
	4. IT Department (System Maintenance)
	5. Suppliers / Contractors (access to internal standards and requirements)
	When Context (Time)
	1. When new documents are created
	2. When old documents are retired
	3. When revisions are made to documents
	4. When documents are accessed
	5. When documents are shared
	Why Context (Motivations)
	1. Maintain Document Integrity
	2. Increase efficiency in storage and retrieval
	3. Promote collaboration
	4. Decrease Ambiguity around revisions
	5. Ensure correct and up to date information is shared

2. Use Cases:

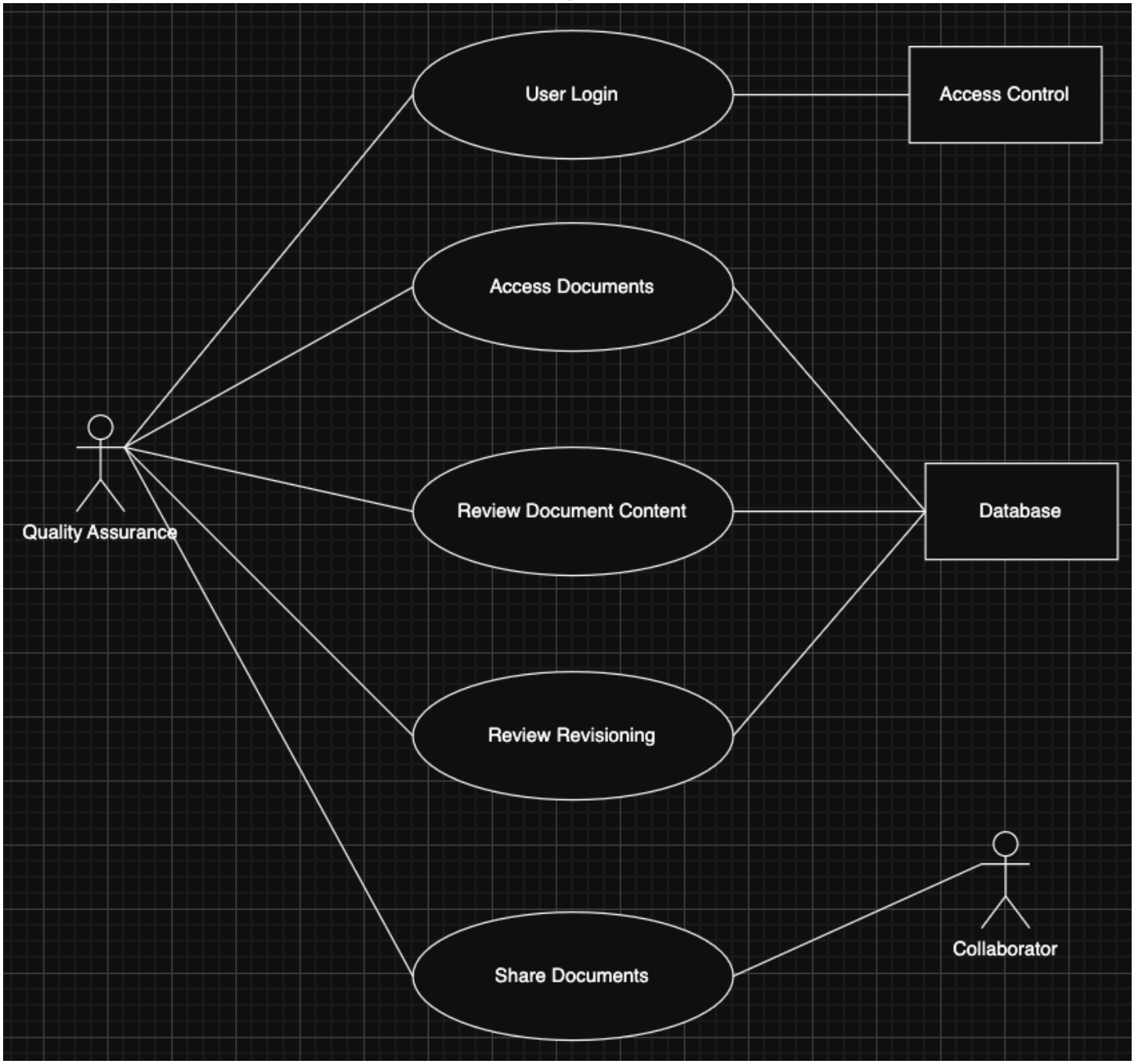
a. 5 Use Case context diagrams

1. Document Owner Use Case Diagram

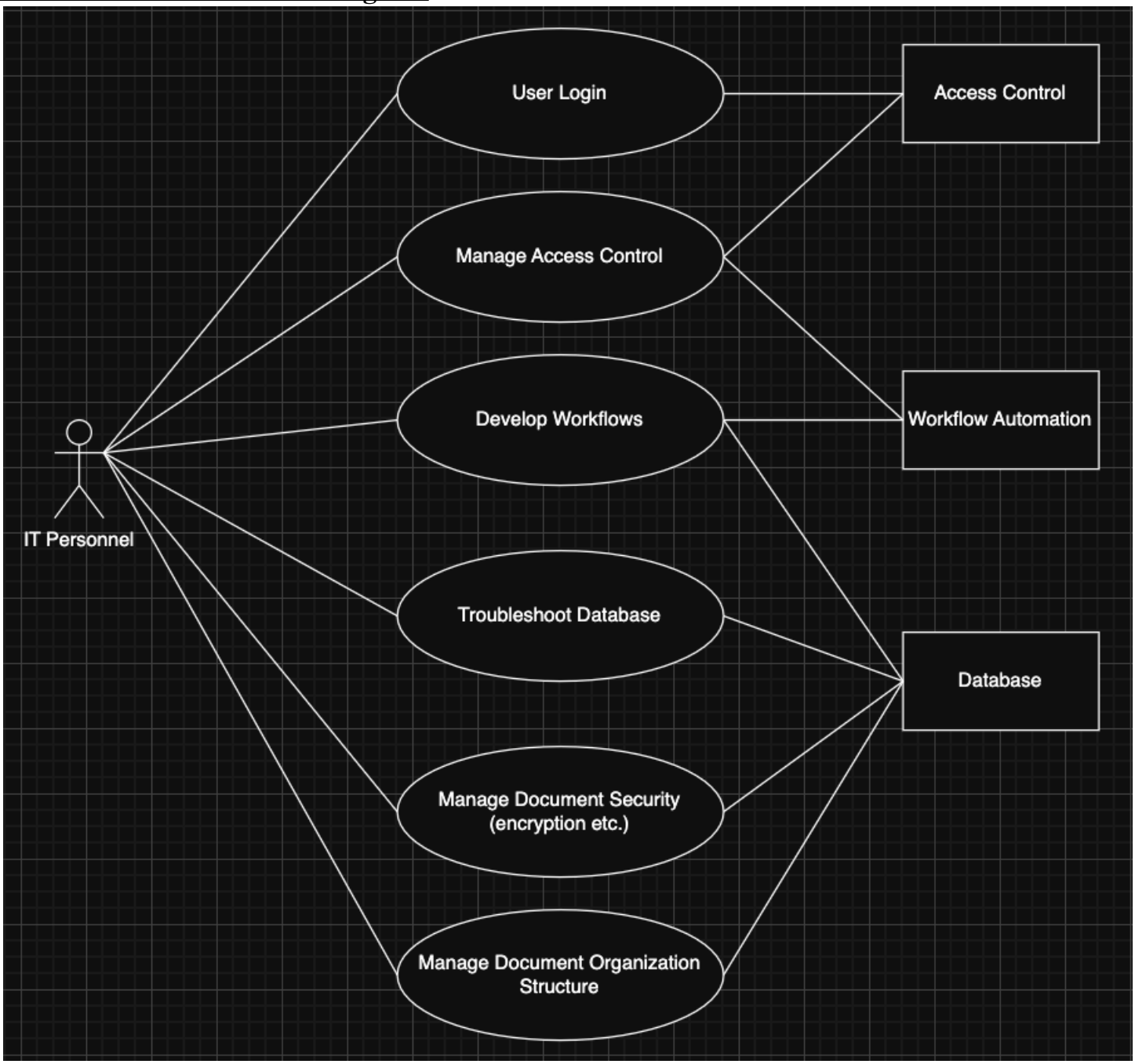
2. Manager Use Case Diagram



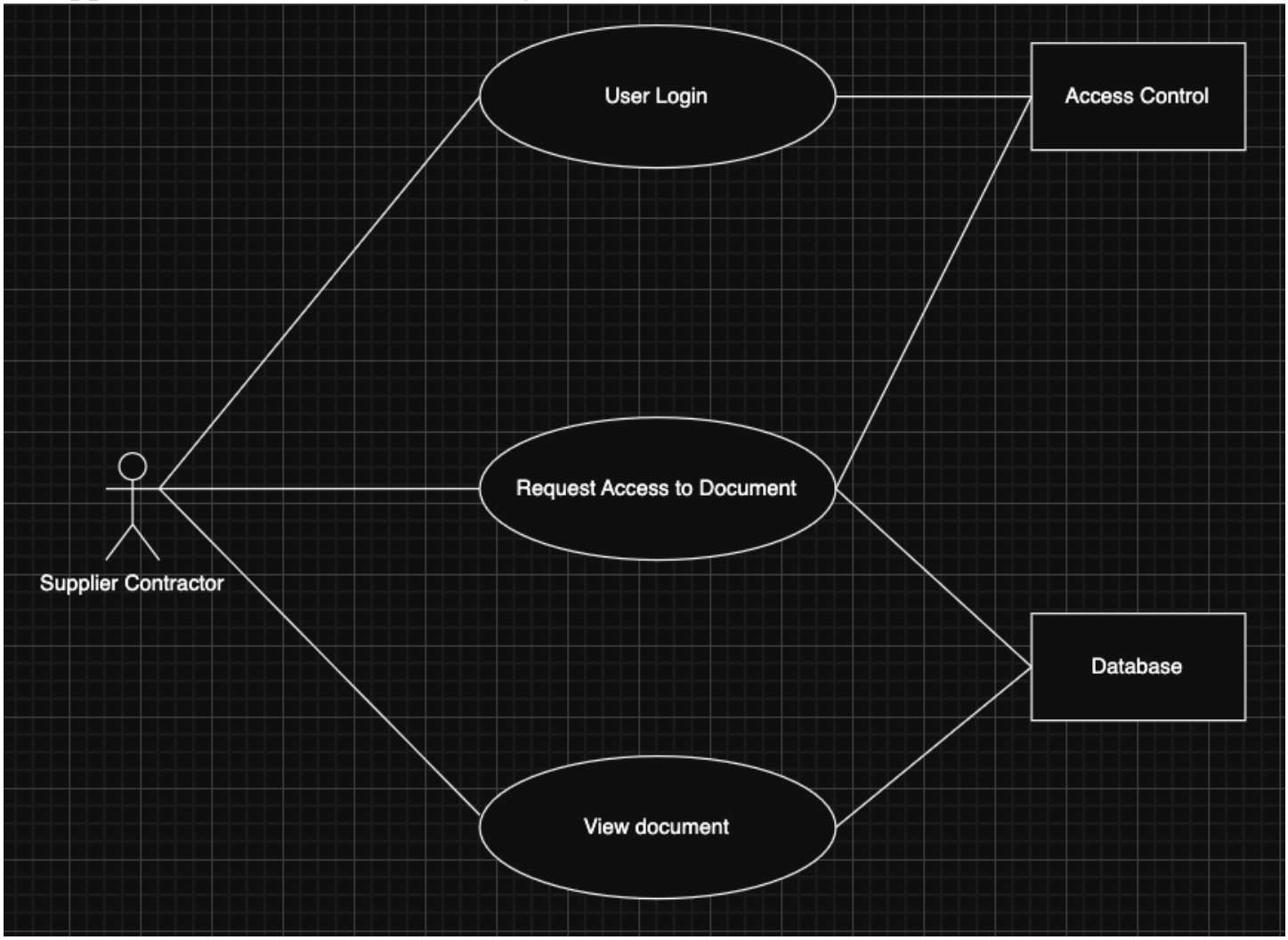
3. Quality Assurance Personnel Use Case Diagram



4. IT Personnel Use Case Diagram



5. Supplier-Contractor Use Case Diagram



b. 5 Narrative Use Cases

Create Document

Use Case Name:	Create Document
Primary Actor:	Document Owner Employee
Description	
1. Employee Logs into System	
2. Access Control Module evaluates users Access Level and grants permissions	
3. User selects the document type they would like to create	
4. System prompts user to input metadata around document	
5. User inputs metadata for document	
6. Workflow creates instance in Database	
7. Employee inputs information into document and saves	
8. Workflow notifies Manager that new document has been requested to be added	
9. Manager Approves document	
10. Wokflow documents revision and commits to database	
11. Access control Module assigns permissions	
11. Employee is notified of document approval and creation	

Approve Revision

Use Case Name:	Approve Revision
Primary Actor:	Manager
Description	
1. Workflow informs by email that a revision to a document has been requested	
2. Manager logs into system	
2. Access Control Module evaluates users Access Level and grants permissions	
3. Manager UI shows revisions requested	
4. Manager Selects specific revision request	
5. System shows synopsis of revision content	
6. Manager reviews change and approves revision	
7. Workflow documents revision and commits to database	
11. Employee is notified of document approval and creation	

Review Documents and Process

Use Case Name:	Review Documents and Process
Primary Actor:	Quality Assurance Personnel
Description	
1. Quality Assurance Employee logs into system	
2. Access Control Module evaluates users Access Level and grants permissions	
3. Quality User selects the document type they would like to review	
4. System gets document and presents to User	
5. Quality User reviews document and revision history for accuracy and ILM purposes	
6. Quality user submits suggestions for revisions	
7. Workflow notifies document owner of request through email	
8. Owner of Document updates the document to request and submits	
9. Workflow notifies manager of submit request	
10. Manager approves revision	
10. Wokflow documents revision and commits to database	
11. Employee is notified of document approval	
12. Quality User is notified of revision approval and commit	

Manage Access Control

Use Case Name:	Manage Access Control
Primary Actor:	IT Personnel
Description	
1. Employee requests enhanced access permissions	
2. Workflow notifies Manager of request	
3. Manager approves request	
4. Workflow Notifies IT Personnel of approved request	
5. IT employee logs into system	
6. Access Control Module evaluates users Access Level and grants permissions	
7. IT employee reviews approval and raises employees access permissions	
8. Access control upgrades employees permissions	
9. Workflow notifies Manager and Employee of completed operation by email	

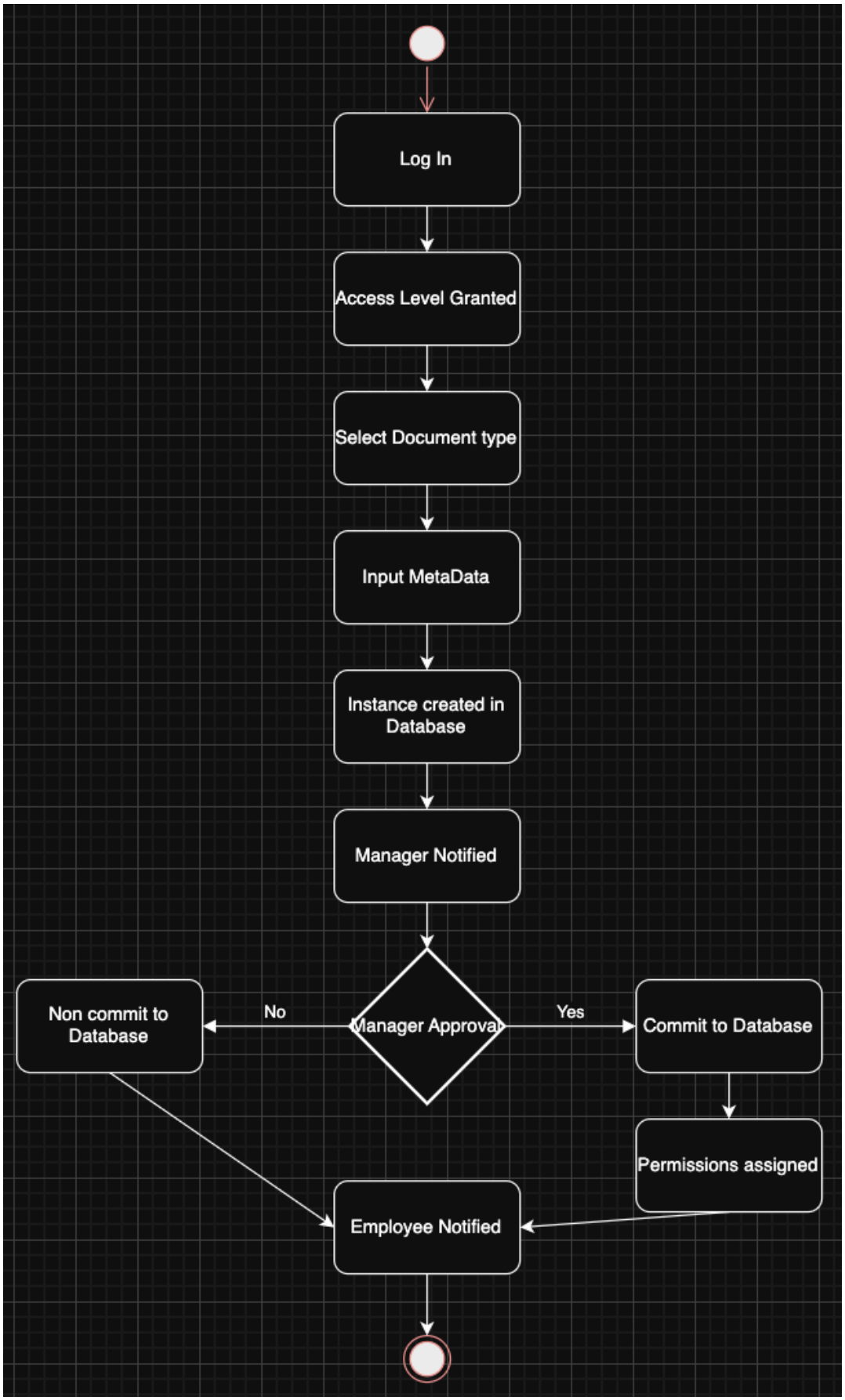
Contractor Request to view document

Use Case Name:	Contractor Request to view document		
Primary Actor:	Supplier / Contractor		
Description			
1. Contractor Logs into system			
2. Access Control Module evaluates users Access Level and grants permissions			
3. Contractor selects document from database they would like to see			
4. Workflow send email to document owner of Contractor Request			
5. Document Owner reviews request and approves			
6. Access Control Module elevates Contractors access to include that documet			
7. Workflow notifies Contractor of elevated access			
8. Contractor opens and views document.			

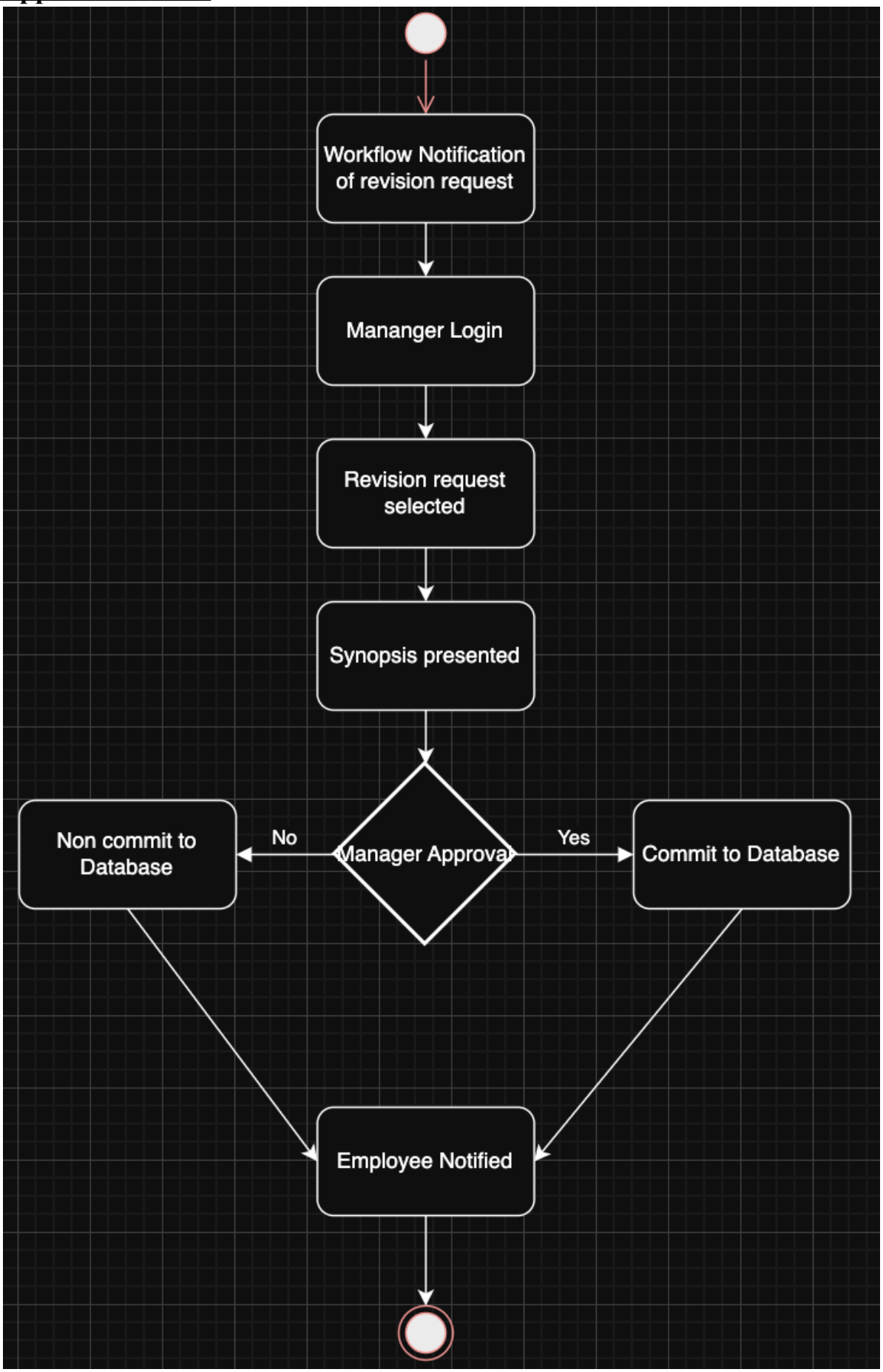
3. Activity and Swim Lane Diagrams:

a. Activity Diagram for each of the Use Case/Narrative Diagrams

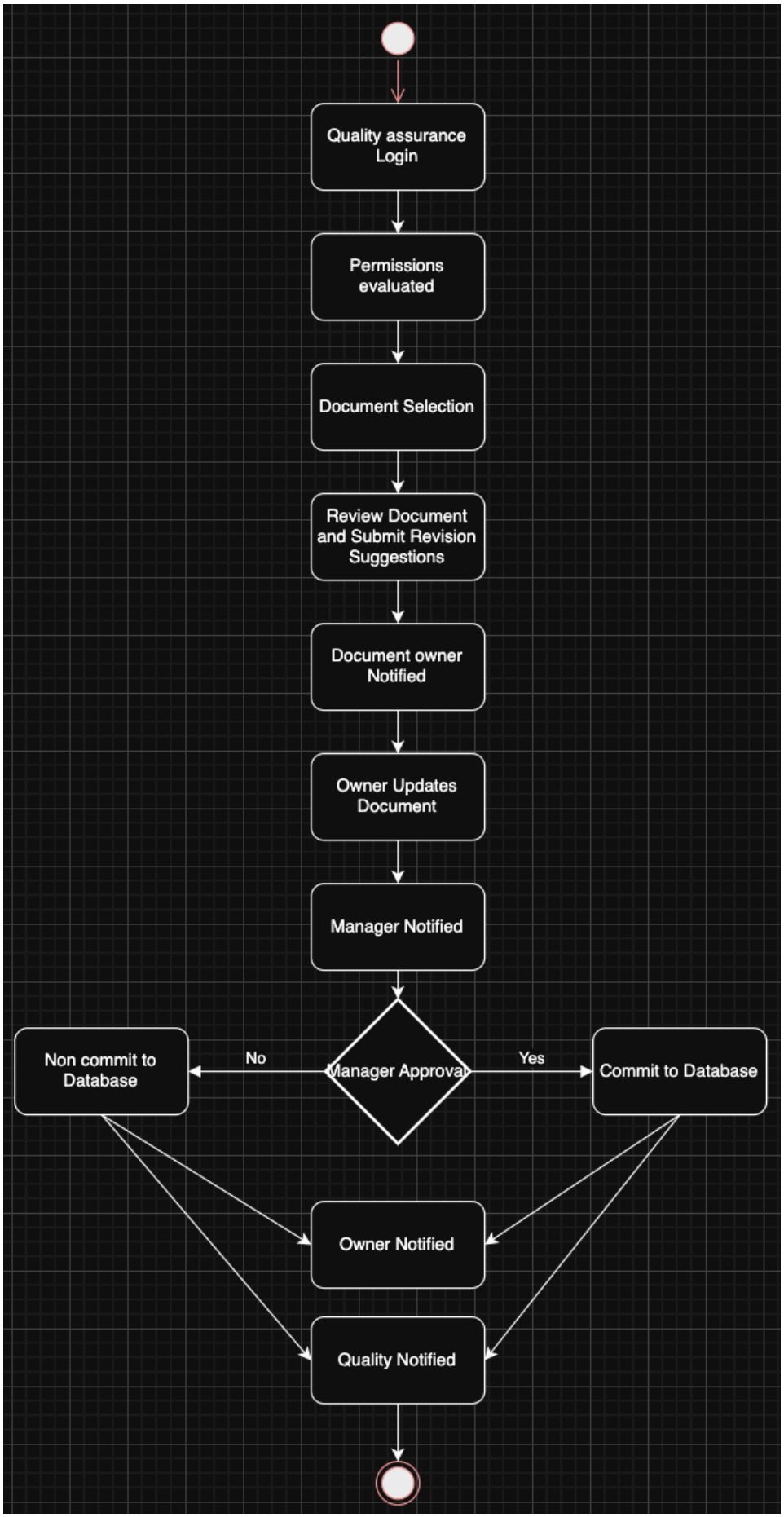
Create Document



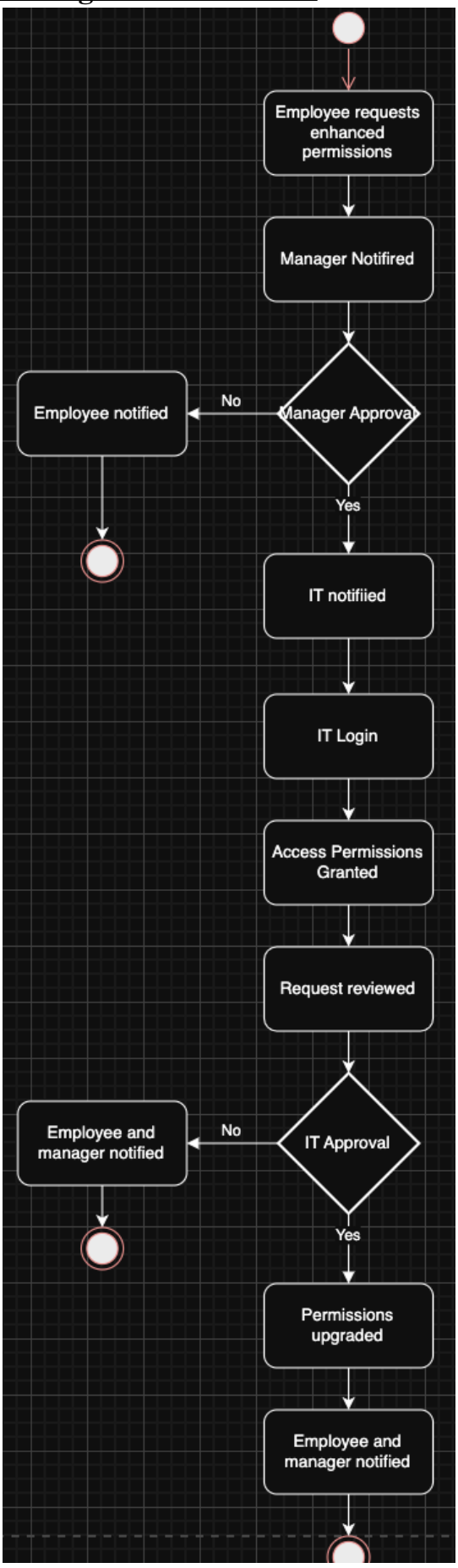
Approve Revision



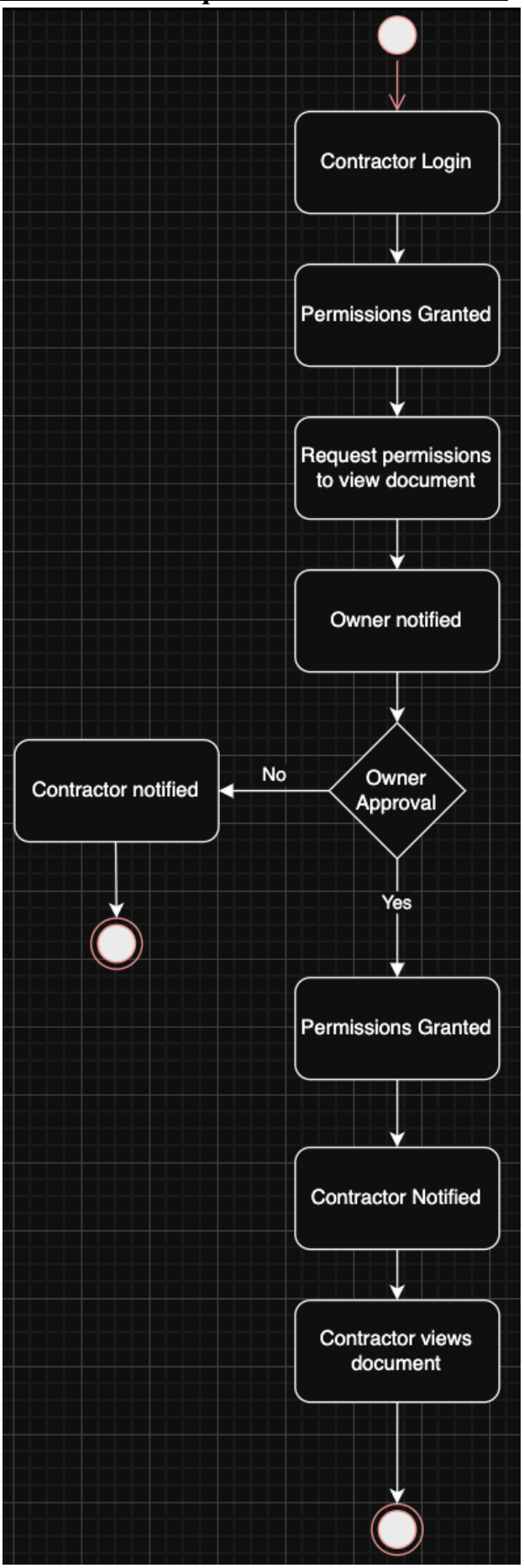
Review Documents and Process



Manage Access Control

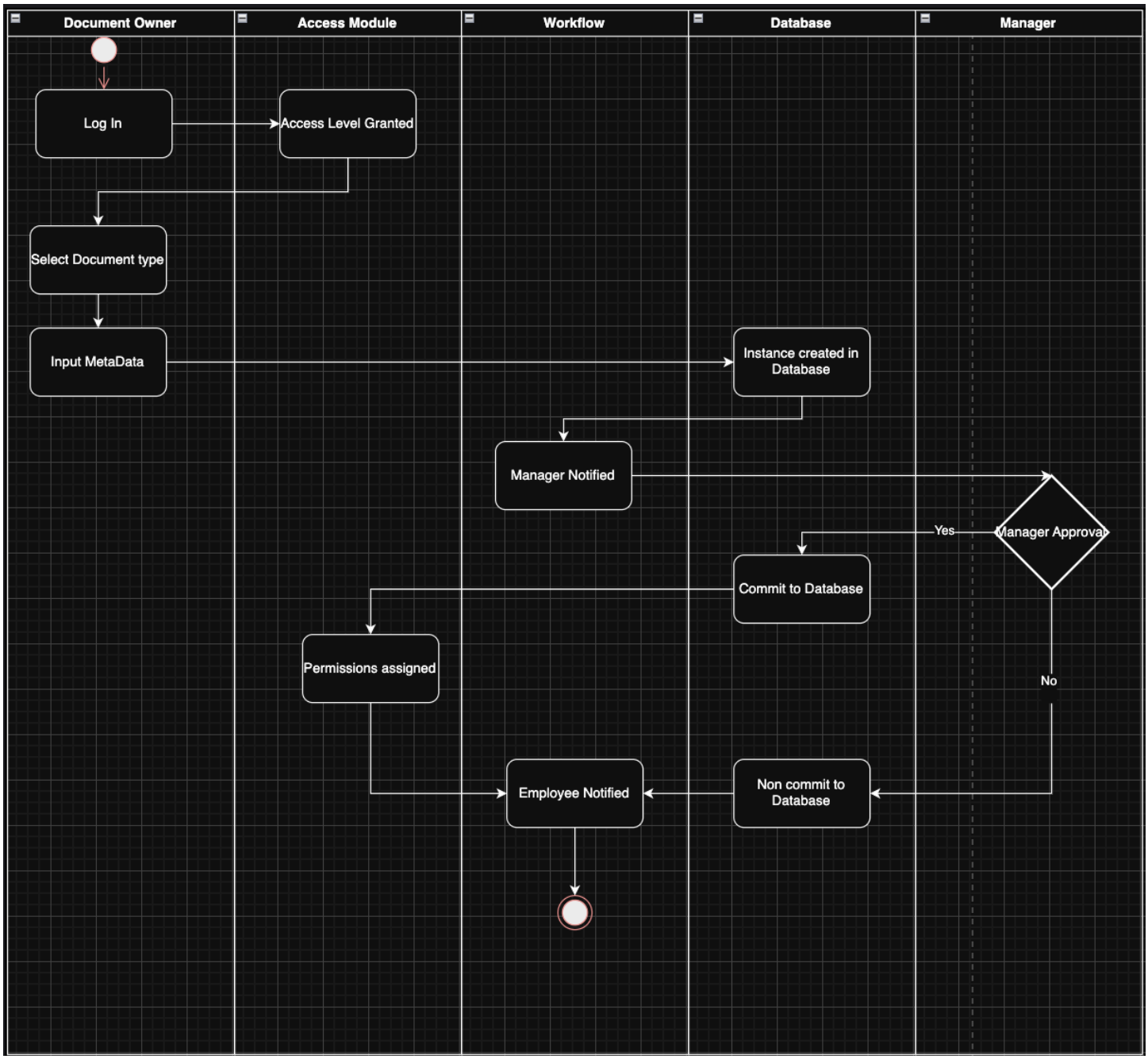


Contractor Request to view document

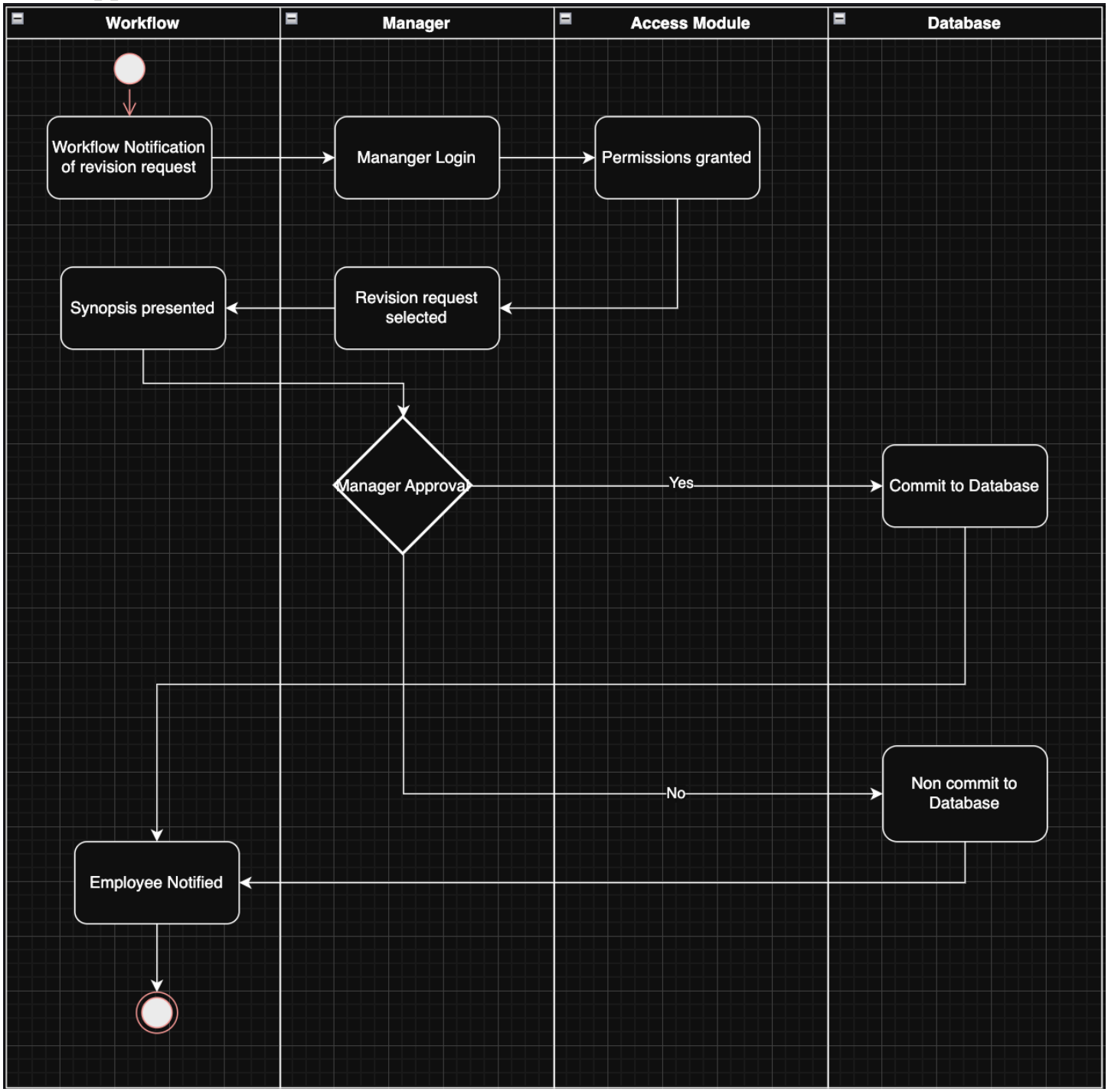


b. Convert each Activity Diagram, from above, into a Swim Lane diagram.

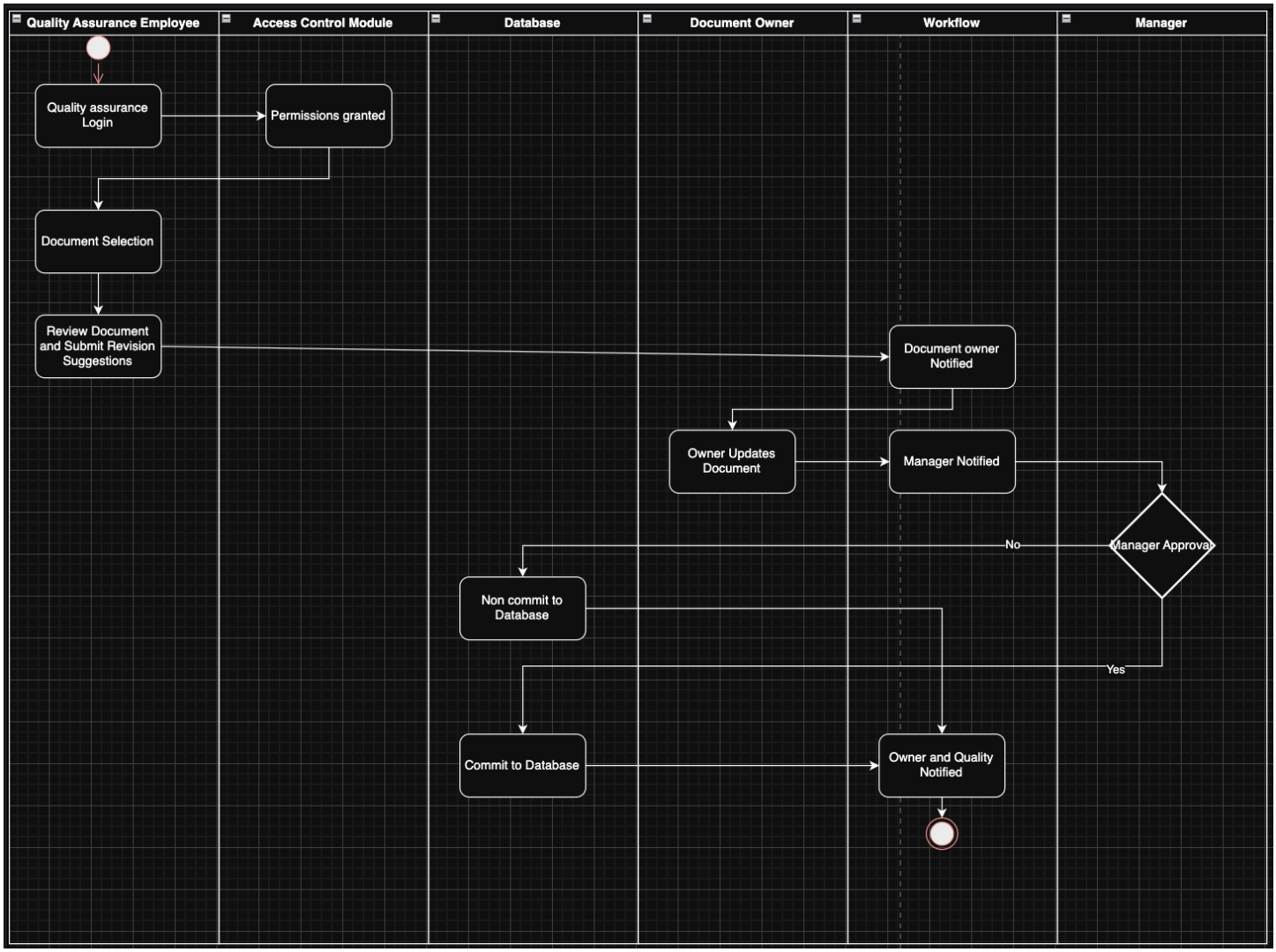
Create Document



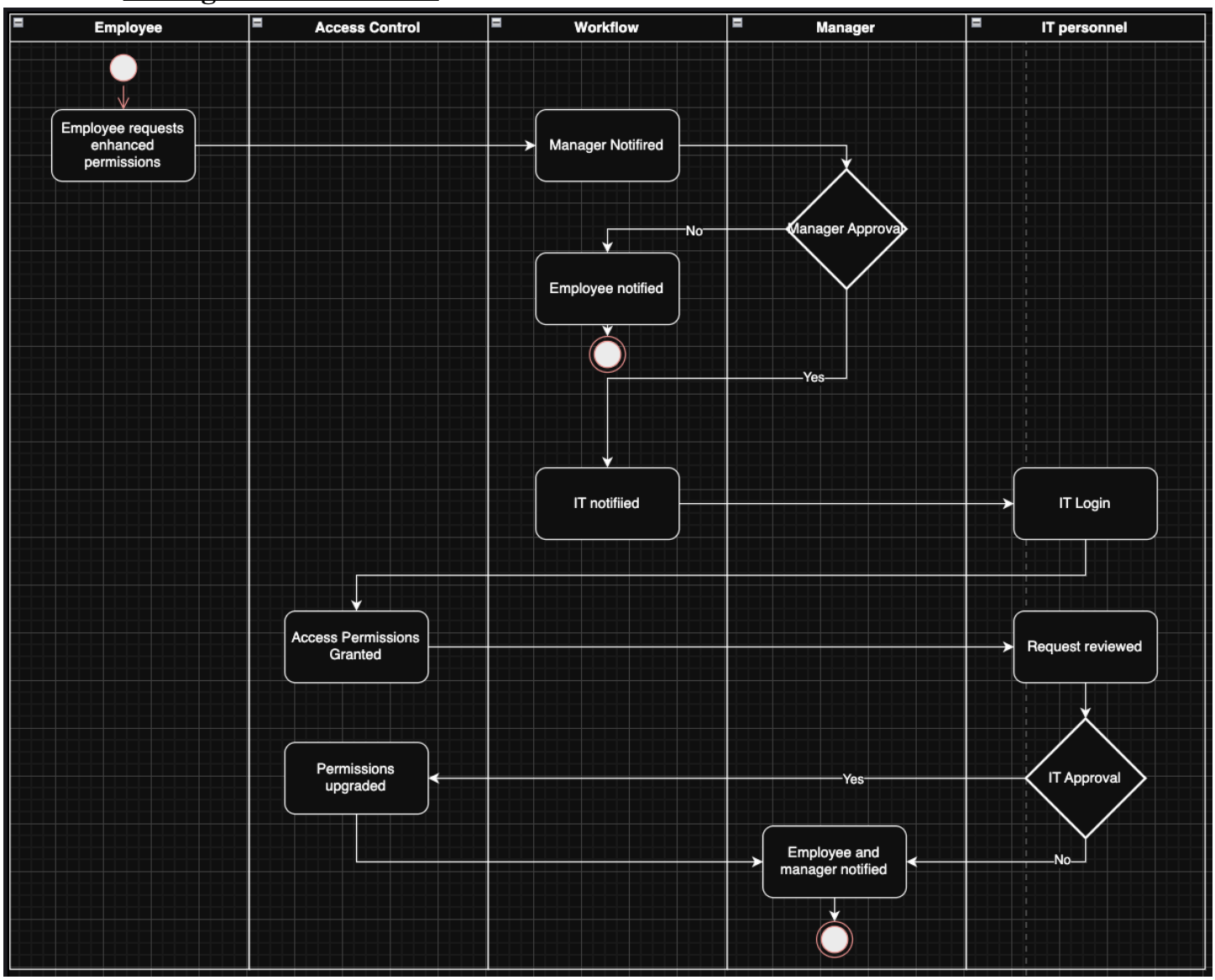
Approve Revision



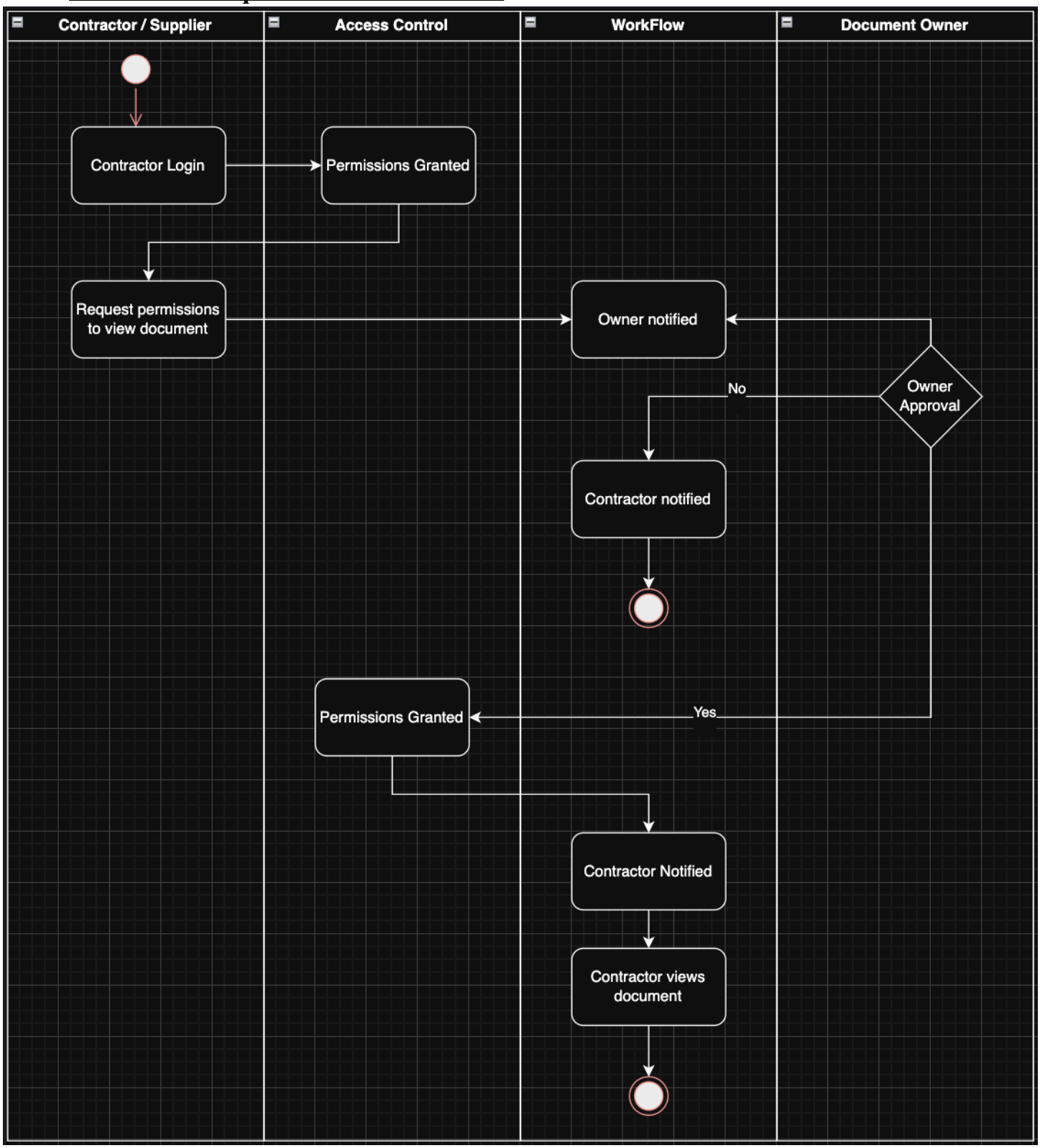
Review Documents and Process



Manage Access Control



Contractor Request to view document



4. Candidate Classes and CRC Cards:

a. 20 candidate application classes

Candidate Classes

Candidate Class List			
Sterotypes			
Actor	**Interface**	**Business**	**Report**
Contractor / Supplier	Login Screen	Permissions	Workflows
IT personnel	Admin Screen	Revision	Documents
Document Owner	Home Screen	Quality	Storage Structure
Manager	Database Interface	Security	Database
Quality assurance employee	Request Screen	Heiarchy	Notifications

b. 10 CRC

Document Owner

Candidate Name:	Document Owner	Stereotype:	Actor
Responsibilities		Collaborators	
Logs into system		All Actors using system	
manages access to owned documents		Database interface	
views documents in database		Database	
Shares documents with others		Workflows	
requests access to documents		Documents	
requests revisions to documents		Notifications	
Purpose / Definition			
The Document owner can create, revise, manage access, and review documents in the system. The Document owner inputs document metadata. The document owner creates documents which engages workflows to notify the appropriate actors to commit data to the database. The Document owner can revise documents which engage workflows to notify appropriate actors and commit documented changes to documents.			

Manager

Candidate Name:	Manager	Stereotype:	Actor
Responsibilities		Collaborators	
Approve Document creation		QA / Document Owner / IT Personnel	
Approve Document Revision		Log in Screen	
Approve elevated access requests		Permissions	
view documents		Database interface	
create documents		Notifications	
		heiarchy	
		Security	
		Documents	
Purpose / Definition			
The Manager class is mainly responsible for responding to requests routed from workflows for various operations. Manager Approval or denial can impact revisions, document creation, or enhanced permissions within the site. Managers may also create document and be document owners.			

Login Screen

Candidate Name:	Login Screen	Stereotype:	Interface
Responsibilities		Collaborators	
Prompts user to login with credentials		All Actors	
invokes access control module with credentials as argument		Security	
Validates Credentials exist		Heiarchy	
Returns appropriate screen to user upon successful login		Permissions	
		home screen	
		Admin Screen	
Purpose / Definition			
The Login screen is the main entry into the system. All actors must engage with the Login Screen in order for the access control module to verify permissions. The login screen passes credentials as an argument to the access control module.			

Database interface

Candidate Name:	Database Interface	Stereotype:	Interface
Responibilities		Collaborators	
Display Documents		All Actors	
Display storage Structure		Login Screen	
Modify document metadata		Home Screen	
Access documents		Permissions	
		Documents	
		Security	
Purpose / Definition			
The Database interface presents database information in a huma friendly manner. It allows an Actor to manipulate and access information stored in the database in a clear and concise manner. The Database interface allows users to request access permissions from document owners and managers and ensures actions performed follow permissions from access control module.			

Revision

Candidate Name:	Revision	Stereotype:	Business
Responsibilities		Collaborators	
Tracks changes to documents		All Actors	
provides document history		Database Interface	
allows reroll to earlier revision		Request Screen	
		Documents	
		Quality	
		Storage Structure	
		Database	
Purpose / Definition			
A revision is an abstract concept that the content of a document has changed enough to require it to be redefined as an entirely new rendition of the document. Quality practices may decide what constitutes a revision or not. A revision engages workflows in the system to send notification to document owners and managers to commit changes to the database. The revisions allow actors to view document history in the Database interface.			

Permissions

Candidate Name:	Permissions	Stereotype:	Business
Responsibilities		Collaborators	
Grants access to various part of system		All Actors	
Ensures Actors only view appropriate database information		Login Screen	
Ensure Actors only enact appropriate actions on the database		Database interface	
directly relates to heiarchy of organization		Revisions	
Helps Determine sequential actions of workflows		Security	
		Workflows	
		Database	
		Notifications	
Purpose / Definition			
Permissions are reviewed and granted after an actors credentials invoke the access control module after logging in. The permissions make sure that the Actor using the system is only able to do what they should be able to do without impacting document integrity or changing database integrity. Permissions also drive the checks and balances that are enacted by workflows.			

Document

Candidate Name:	Document	Stereotype:	Report
Responsibilities		Collaborators	
Store metadata / information		All Actors	
Add Value to business		database interface	
		request screen	
		permissions	
		Quality	
		Security	
		Workflows	
		Database	
Purpose / Definition			
Documents have the simple job of housing information. These are pieces of "work" that are commited to the database. The whole system is developed to manage this singular piece of "work". These documents have information that directly adds value to the business itself.			

Database

Candidate Name:	Database	Stereotype:	Report
Responsibilities		Collaborators	
Houses Documents		All Actors	
Houses Metadata		Database Interface	
Houses Revision Information		Permissions	
		Security	
		Storage Structure	
Purpose / Definition			
The Database acts solely as a repository for documents and the information or metadata attached to them. Most modules within the system interact directly with the information in the database and present it to the Actor. Database must accept commits and track revisions.			

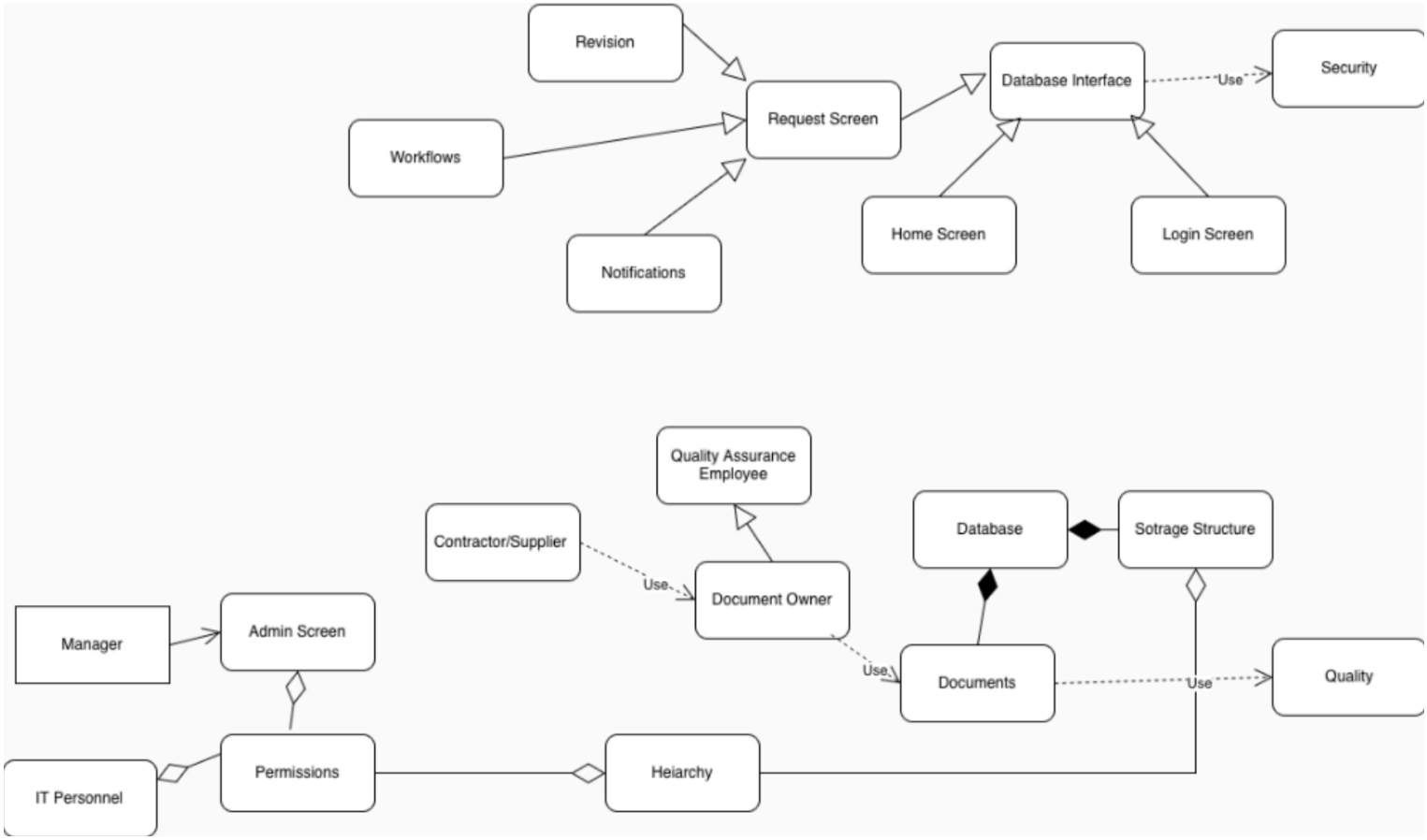
Hiearchy

Candidate Name:	Hiearchy	Stereotype:	Business
Responsibilities		Collaborators	
Helps order how decisions are made		All Actors	
Provides Structure to organization		Login Screen	
Provides Context to workflows		Permissions	
Enables Quality Checks and balances		Quality	
Creates ownership and responsibiliy		Security	
		Workflows	
		Database	
		Notifications	
Purpose / Definition			
Heiarchy is the concept that there are tiered levels of responsibility and ownership within an organization. Heiarchy helps to drive the permissions behind Actors such as Manager and Contractor or Employee. Many Security features will be driven from the concept that there is a pecking order within the organziation and only certain users should be able to make certain decisons.			

Notification

Candidate Name:	Notification	Stereotype:	Report
Responsibilities		Collaborators	
Alerts Actors when action is required		All Actors	
Alerts Actors when actions are completed		Permissions	
		Revision	
		Quality	
		Security	
		Workflows	
		Database	
		Heiarchy	
Purpose / Definition			
A Notifcation is meant to alert and Actor when a action is either required or has taken place. It is generally give when a response to a request is fullfilled or a request has been generated by an Actor. Without a notification there wouldn't be any urgency around processing Database work.			

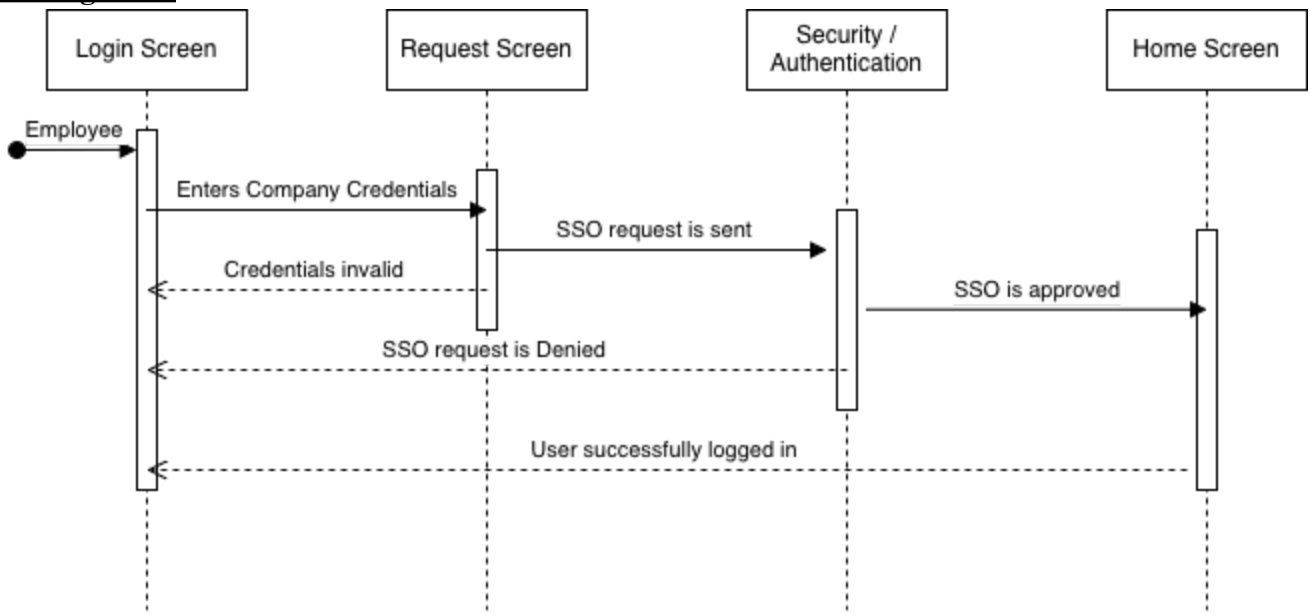
5. Class Relationships Diagrams



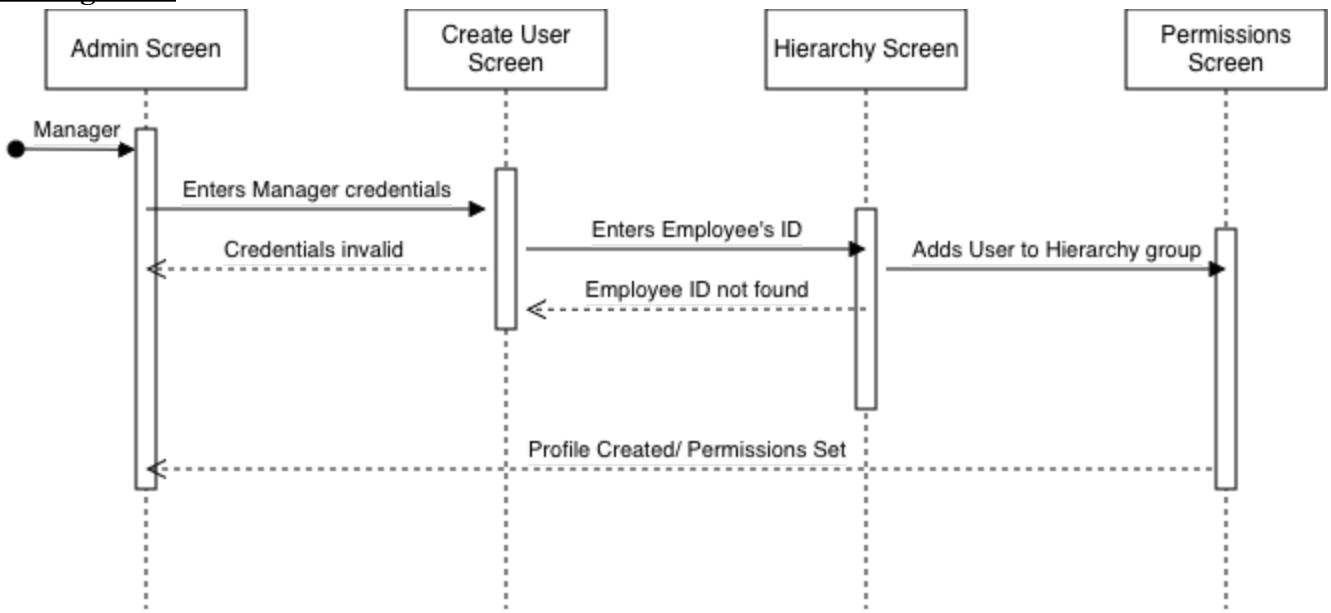
6. Sequence Diagram/Charts and State Diagrams:

a. Sequence Diagrams/Charts

Sequence Diagram 1

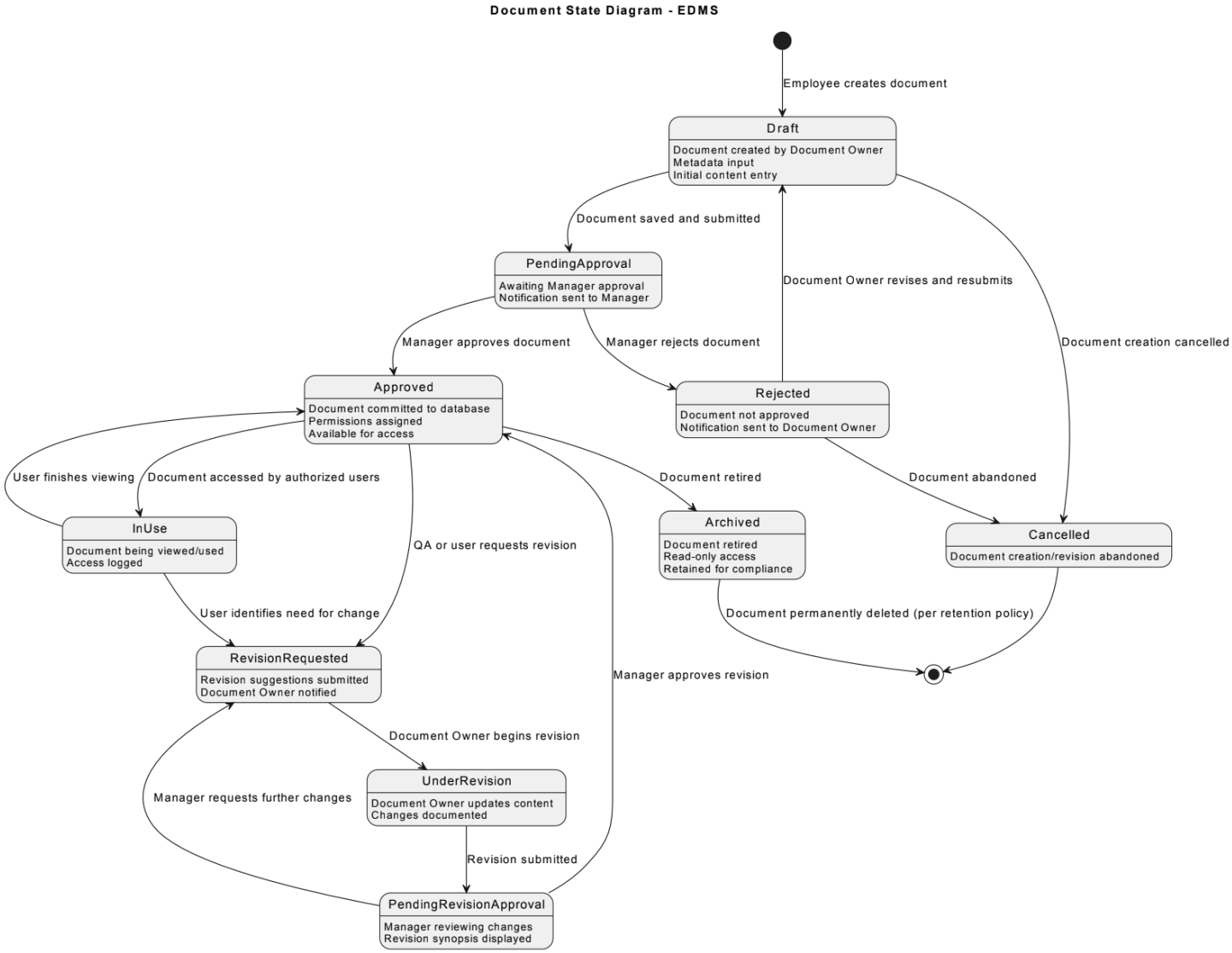


Sequence Diagram 2

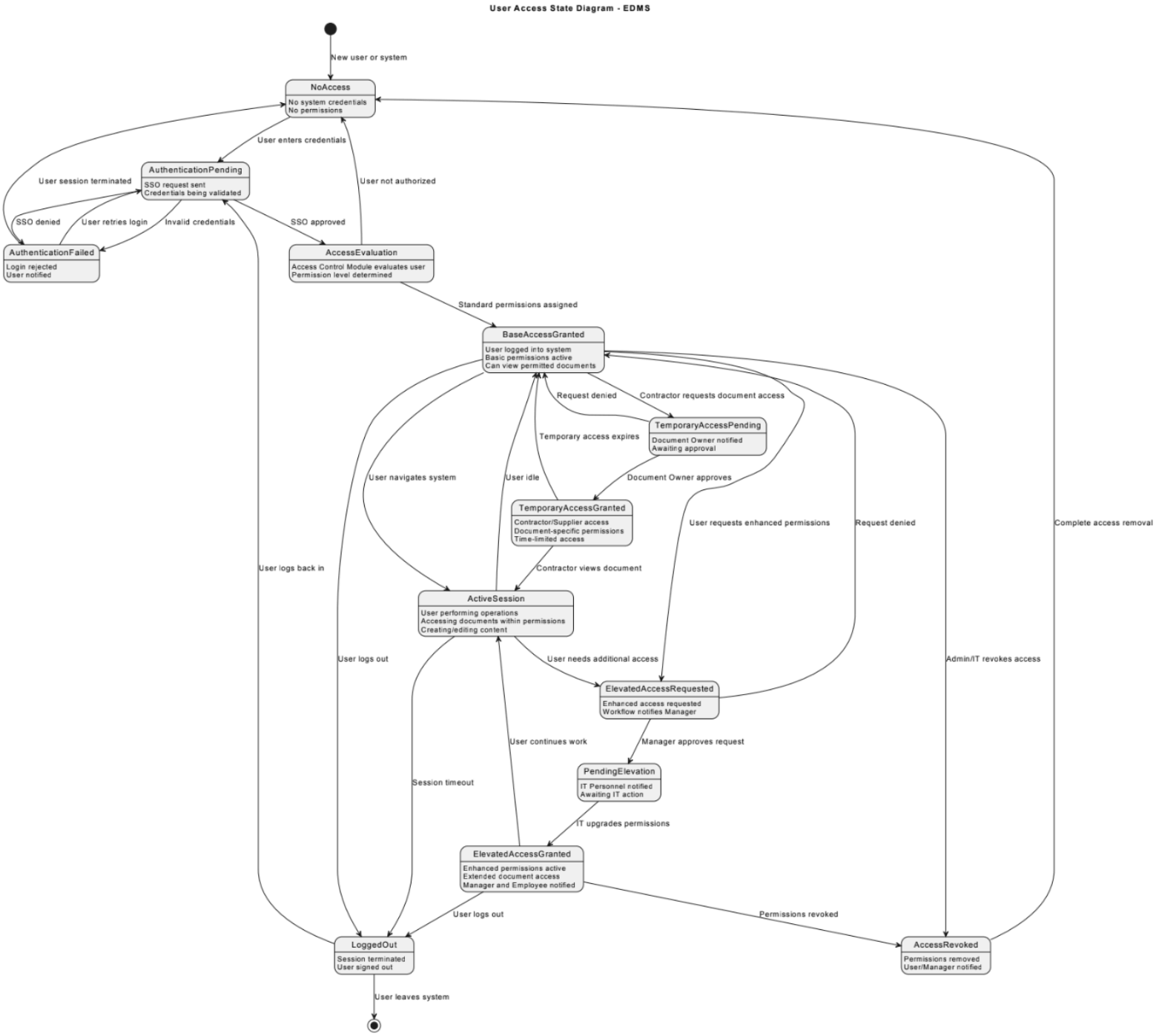


b. State Diagrams

Document State Diagram



User Access State Diagram



7. Midterm Group Evaluation: The team collaborated effectively to accomplish the objectives outlined in this midterm exam. Since the exam's release, we met regularly to discuss our progress, share insights, and resolve challenges together. Each member contributed their unique tools and working styles, and we divided the tasks evenly across the group. By organizing our materials in Google Drive with dedicated folders for each assignment, we maintained clear workflows and efficiently assembled this final document.